

NeuroScan AI

User Guide for Clinical Staff

Brain MRI classification support tool. Prepared for neurosurgeons, radiologists, and clinical reviewers.

Read this before first use

This tool provides decision support only. It does not diagnose. Every result must be reviewed by a licensed radiologist or neurosurgeon before any clinical action is taken.

1. What This Application Does

NeuroScan AI classifies a brain MRI scan into one of four categories: Glioma, Meningioma, Pituitary Tumor, or No Tumor. It uses a trained convolutional neural network (CNN), shows a Grad-CAM heatmap of the regions the model focused on, and produces a written clinical report describing the result.

The application has two distinct sources of output, and it is important to know the difference:

- Prediction, confidence score, and heatmap: generated entirely by the trained CNN classifier.
- Written clinical report text: drafted by Claude, an AI language model from Anthropic, based only on the classifier's numeric output. Claude does not view the image and does not make the classification.

2. Step by Step: Running an Analysis

1. Open the application in your browser.
2. In the left panel, confirm Model Mode. MobileNetV2 only is faster. Use the full ensemble if available for higher reliability.
3. Upload an MRI image (JPG, PNG, or BMP, under 10 MB) using the Upload button, or select a sample image from the dropdown.
4. The image is automatically validated as a brain MRI. If validation fails, review the reason shown before proceeding. An override option exists but should only be used when you are confident the image is a valid scan.
5. The application runs four steps automatically: validation, AI classification, Grad-CAM heatmap generation, and clinical report drafting. A loading indicator shows which step is in progress.
6. Review the prediction, confidence percentage, and class distribution bars.
7. Review the Grad-CAM heatmap and its color explanation (see Section 4 below).
8. Review the Clinical Report section, using the Findings, Reasoning, Patient Summary, and Reliability tabs.
9. Confirm all findings against the actual imaging and patient context before making any clinical decision.

3. Understanding the Prediction and Confidence

The confidence percentage reflects how strongly the model favors its top prediction relative to the other three classes. It is a statistical output of the model, not a probability of clinical certainty.

When more than one model is loaded (ensemble mode), individual model votes are shown before averaging. Agreement between models supports the result. Disagreement between models is a signal to weigh the result more cautiously and prioritize direct radiologist review.

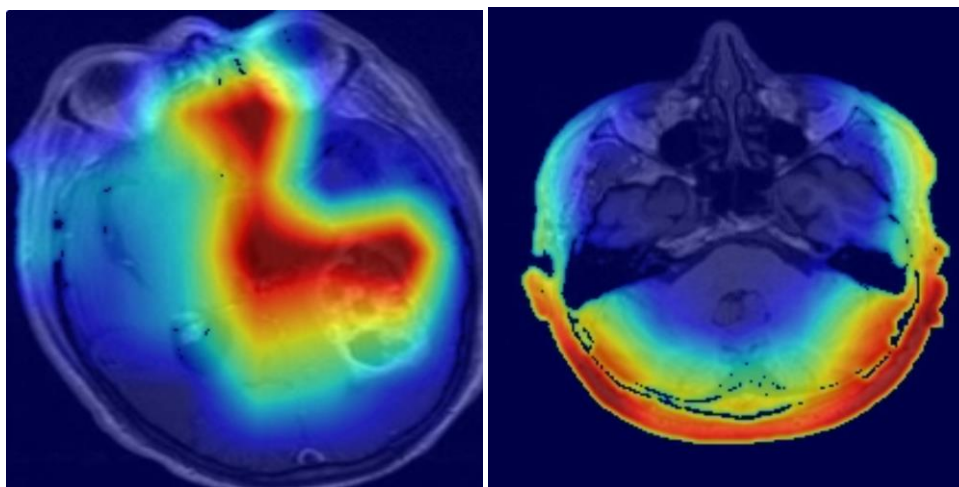
Low confidence results

A low confidence score does not rule out disease. It means the classifier itself is uncertain. Treat any low confidence result as inconclusive and defer to direct image review.

4. Understanding the Grad-CAM Heatmap

Grad-CAM shows which regions of the image the classifier weighted most heavily to reach its specific prediction. It is a visualization of model attention, not a segmented tumor boundary and not a radiologist's localization of disease.

Color	Meaning
Blue / low activation	Little to no influence on the model's decision.
Cyan to yellow	Moderate influence on the model's decision.
Orange to red	High influence on the model's decision.



Important: Red does not automatically mean "there is a tumor here."

For a **No Tumor** prediction, red or orange regions often highlight normal anatomical structures, such as the skull, scalp, ventricles, or symmetric brain tissue, that the model found informative for identifying a normal scan. These highlighted areas should not be interpreted as suspected lesions.

For a **Tumor** prediction, concentrated activation over anatomically plausible regions may indicate areas that contributed strongly to the model's decision. However, these heatmaps do **not** precisely delineate the tumor and should not be interpreted as lesion segmentation or a substitute for expert radiological assessment of the original MRI.

5. Understanding the Clinical Report

The Clinical Report section is written by Claude (Anthropic) using only the classifier's numeric output: predicted class, confidence, per-model agreement, Grad-CAM statistics, and computed image quality. A visible note above the report confirms this every time a report is generated.

The report is organized into four tabs:

- Findings: predicted class, location and morphology notes, and image quality.
- Reasoning: how the model reached its result and what the Grad-CAM activation pattern shows.
- Patient Summary: a plain language explanation suitable for sharing context with a patient, alongside clinical judgment.
- Reliability: a computed reliability score, overall reliability phrase, uncertainty factors, and differential diagnosis considerations.

If Claude is unavailable, the application shows a fixed template report instead of a per-image one, and displays a clear notice explaining why (for example, an account credit or connectivity issue). This notice is a system status message, not a defect in the analysis already shown above.

6. Image Quality

Image quality is computed automatically from pixel contrast and dynamic range and shown as GOOD, ADEQUATE, or POOR. A POOR rating means the underlying image itself may limit how much weight should be placed on the result, independent of the confidence score.

7. Limitations

- This tool is a classification aid. It is not a diagnostic device and has not been validated for autonomous clinical decision making.
- It classifies into four categories only. It cannot detect conditions outside those categories.
- It does not localize a lesion with clinical precision. Anatomical location must be confirmed by direct radiologist review of the imaging.
- Model confidence and image quality are statistical outputs, not clinical certainty.
- All results, in every case, require confirmation by a licensed radiologist or neurosurgeon reviewing the actual imaging and patient history.

8. Dataset and Model Performance

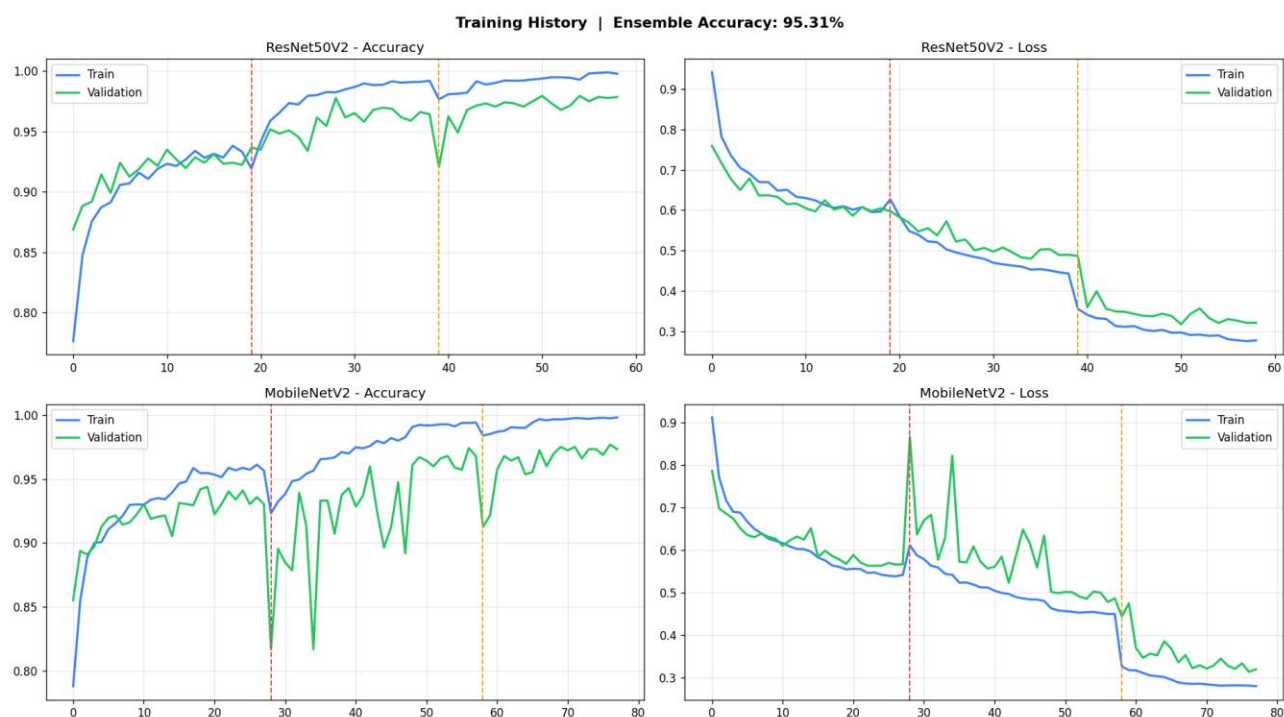
This section documents what the underlying models were trained and evaluated on. It is provided for transparency, not as evidence of clinical validation.

✓ Dataset

The classifiers were trained on the Brain Tumor MRI Dataset published on Kaggle by Masoud Nickparvar, approximately 7,000 MRI images across four classes: Glioma, Meningioma, No Tumor, and Pituitary Tumor.

✓ Training Results

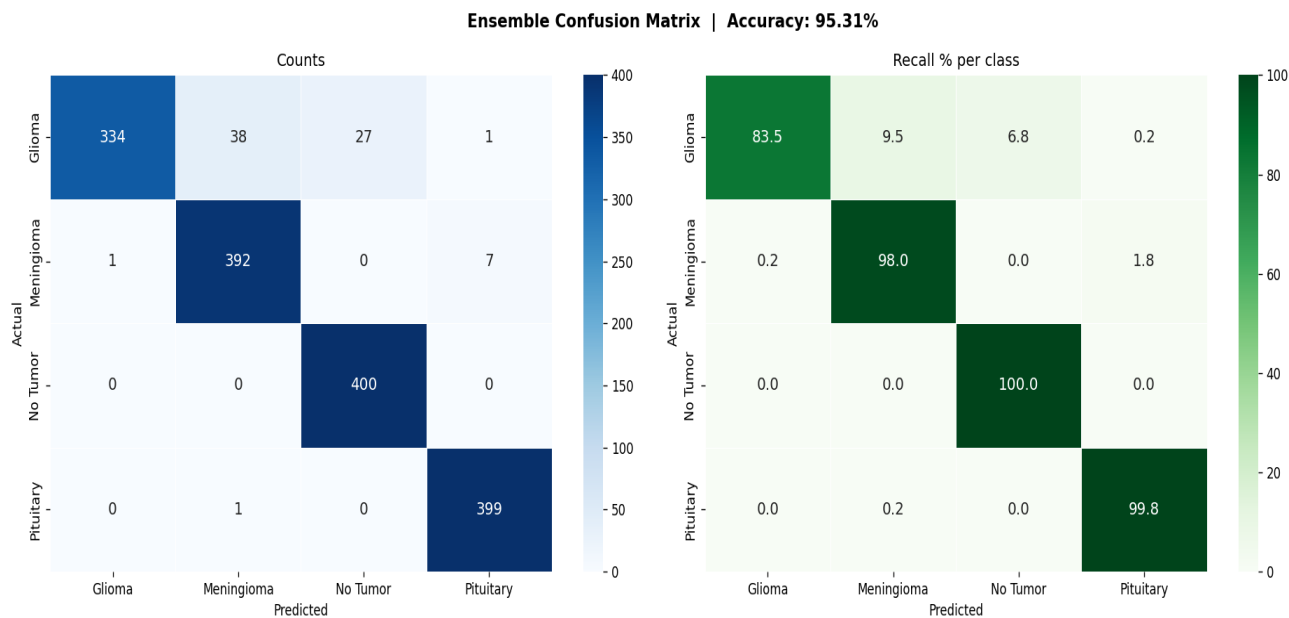
Two convolutional neural networks were trained: ResNet50V2 and MobileNetV2. Both were fine-tuned in stages, shown by the vertical markers in the chart below, where additional layers were progressively unfrozen. Training accuracy, in blue, and validation accuracy, in green, both increased over training and converged by the end of training for each model.



Training and validation accuracy and loss for ResNet50V2 and MobileNetV2. Ensemble accuracy 95.31%.

✓ Test Set Performance (Ensemble)

The combined ensemble was evaluated on a held-out test set not used during training, reaching 95.31% overall accuracy. The confusion matrix and per-class recall are shown below.



Ensemble confusion matrix and per-class recall on the held-out test set (1,600 images, 400 per class).

Class	Recall on test set
Glioma	83.5%
Meningioma	98.0%
No Tumor	100.0%
Pituitary Tumor	99.8%

What this data does and does not show

These figures describe performance on the internal held-out test set from the same dataset used for training. They are not a substitute for independent clinical validation on external patient populations.

Glioma had the lowest recall of the four classes, meaning it was the class most often confused with another category during testing. Extra caution is warranted when the predicted class is Glioma or when Glioma is a close second in the class distribution.

9. Quick Reference

Situation	What to do
Confidence below roughly 60%	Treat as inconclusive. Prioritize direct image review.
Models disagree in ensemble mode	Weigh the result cautiously. Do not rely on the average alone.
Image quality is POOR	Consider re-scanning or re-uploading a higher quality image before relying on the result.
Static template report shown	Read the notice above the report for the reason. It is a system status, not an error in the prediction above.
MRI validation fails	Confirm the file is an actual brain MRI before using the override option.

Summary

NeuroScan AI supports review. It does not replace clinical judgment. Every prediction, heatmap, and report requires confirmation by a licensed radiologist or neurosurgeon before any decision is made.

NeuroScan AI - <https://brain-tumor-prediction-mri-classifier.streamlit.app/>